GLOBAL SCHOOL OF SCIENCE

Pre-Board Examination - 2079

Class : XII | **Subject: Chemistry (3021)** | **Full Marks: 75**

Time: 3 hrs. | **SET 'B'** | **Pass Marks: 30**

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

1. 0.4 gram NaOH is added to 10 ml of 1N HCl, the resulting solution is
   a) Acidic b) Alkaline
   c) Neutral d) None
2. Value of ionic product of water at 393 K is
   a) Less than 1×10^{-14} b) Greater than 1×10^{-14}
   c) Equal to 1×10^{-15} d) Equal to 1×10^{-7}
3. Half life of a first order reaction is 2.31 min. The rate constant of the reaction is
   a) 0.2 min^{-1} b) 0.3 min^{-1}
   c) 2 min^{-1} d) 3 min^{-1}
4. Heat exchanged in a chemical reaction at constant temperature and pressure is called
   a) internal energy b) enthalpy
   c) entropy d) free energy

5. Which of the following metal is leached by cyanide process?

a) Ag b) Na

c) Al d) Cu

6. Which element in d-block shows maximum oxidation state?

a) Zn b) Fe

c) Mn d) Cu

7. Ethanamide is obtained from ethanoic acid by the reaction of

a) Conc. H_2SO_4 b) P_2O_5

c) NH_3 d) Anhy. $AlCl_3$

8. Nitrobenzene is reduced with zinc and sodium hydroxide in methanol. The product is,

a) Aniline b) Azobenzene

c) p-aminophenol d) Hydrazobenzene

9. Acetaldehyde and acetophenone can be distinguish by using the reagent.

a) $NaOH+I_2$ b) 2, 4- DNPH test

c) Fehling's solution d) $NH_2–NH_2$

10. The number of possible structural isomers of 1° amines of molecular formula $C_4H_{11}N$ give

a) 1 b) 2

c) 3 d) 4

11. Acetamide is

a) acidic b) basic c) neutral d) amphoteric

Group - B

Give short answer to the following questions. **[8 × 5 = 40]**

1. Ionic product of water at 25°C is 1×10^{-14} and water is regarded as very weak electrolyte.

   i. Define ionic product of water. [1]

   ii. Deduce the relation $K_w = [H^+] [OH^-]$ [1]

   iii. If equal volume of two solutions having (pH = 1) and (pH = 12) are mixed together, what will be the p^H of the resulting solution? [3]

2. A part of electrochemical series is shown below

Electrode system	Electrode reaction	E° Volt
$Li^+ \mid Li$	$Li^+ + e^- \rightleftharpoons Li$	-3.05
$K^+ \mid K$	$K^+ + e^- \rightleftharpoons K$	-2.925
$Ca^{++} \mid Ca$	$Ca^{++} + 2e^- \rightleftharpoons Ca$	-2.870
$Mg^{++} \mid Mg$	$Mg + 2e^- \rightleftharpoons Mg$	-2.370
$Zn^{++} \mid Zn$	$Zn^{++} + 2e^- \rightleftharpoons Zn$	-0.760
$Fe^{++} \mid Fe$	$Fe^{++} + 2e^- \rightleftharpoons Fe$	-0.440
$Sn^{++} \mid Sn$	$Sn^{++} + 2e^- \rightleftharpoons Sn$	-0.136
$H^+ \mid H_2$	$2H^+ + 2e^- \rightleftharpoons H_2$	0.00
$Sn^{+4} \mid Sn^{+2}$	$Sn^{+4} + 2e^- \rightleftharpoons Sn^{+2}$	+0.15
$Cu^{++} \mid Cu$	$Cu^{++} + 2e^- \rightleftharpoons Cu$	+0.337
$Fe^{+++} \mid Fe^{++}$	$Fe^{+++} + e^- \rightleftharpoons Fe^{++}$	+0.977
$Ag^+ \mid Ag$	$Ag^+ + e^- \rightleftharpoons Ag$	+0.799

a. Define electrochemical series. [1]

b. Write the cell notation for galvanic cell formed by using Mg^{++}/Mg and Fe^{++} / Fe electrode. Which one is anode and which one is cathode? Calculate the cell potential of for the cell. [3]

c. Does the following reaction occur? [1]

$2Fe^{++} + Sn^{4+} \rightarrow 2Fe^{+++} + Sn^{++}$

OR

Distinguish enthalpy of formation and enthalpy of combustion. Calculate the enthalpy of combustion of methane if enthalpy of formation of methane, water and carbondioxide are –17.9, –72 and –93 Kcal respectively. [1+4]

3. a. Write the important features of transition metal. [2]

b. Predict the shape of $[Cu(NH_3)_4(H_2O)_2]^{2+}$ and $[CuCl_4]^{2-}$. [1]

c. Explain with reason:

i. Zn^{++} salts are white while Cu^{++} salts are blue. [1]

ii. Transition metals and many of their compounds acts as a good catalyst. [1]

4. a. Penta-hydrated copper sulphate is called blue vitriol. [2]

i. Starting from metallic copper, how can you obtain blue vitriol?

ii. What happens when aqueous solution of blue vitriol is treated with excess ammonia solution ?

b. A metal (M) is extracted from its suphide ore whose atomic number is 80. It also occurs as the amalgams of certain metals and is popularly known as quick silver. [3]

i. Name the metal 'M' and write down the molecular formula of its common ore.

ii. Why is this metal called quick silver?

iii. Write down the chemical reaction involved during reduction in its extraction.

5. A sweet smelling organic compound (A) slowly oxidized by air in the presence of sunlight to give highly poisonous gas carbonyl chloride.

i. Why above compound (A) is stored in dark coloured bottle?

ii. Give principle reaction involved in the preparation of compound (A) from ethanol.

iii. How would you convert compound (A) into

(a) Hypnotic drug

(b) Chloropicrin [1+2+2]

6. a. Give proper reason [3]

i. Phenol is more acidic than aliphatic alcohol.

ii. Cyanide ion is ambident nucleophile.

iii. Amino group is protected before nitration of aniline.

b. Arrange the following compounds in the order of increasing basic character with proper reason,

$CH_3 - NH_2$, NH_3, $(CH_3)_2\ NH$, $C_6H_5\ NH_2$ [2]

7. Write down the isomeric alcohols of C_3H_8O and their IUPAC name. How would you apply Victor Meyer's test to distinguish these isomers? [2+3]

OR

a. How is phenol obtained from [2]

i. benzenediazonium salt

ii. chlorobenzne

b. Starting from phenol how would you obtain [2]

i. phenolphthalein

ii. Salicyldehyde

c. How is presence of phenolic group in organic compound detected? [1]

8. Write an example of each of the following reactions. [1×5]

i. Hydroboration oxidation

ii. Perkin condensation

iii. Sandmeyer's reaction

iv. Cannizzaro's reaction

v. Claisen condensation

Group - C

Give long answer to the following questions. **[3 × 8 = 24]**

9. a. 1 gm of a divalent metal was dissolved in 25 ml of 1M H_2SO_4. The unreacted acid further required 15 ml of NaOH (f = 0.8) for complete neutralization.

i. Calculate the gram equivalent of unreacted acid

ii. Find the atomic weight of metal. [1+3]

b. There are different theories to describe acids and bases.

i. Define acid and base on the basis of Bronsted – Lowry concept with examples. [2]

ii. What are conjugate acid base pair? Write the conjugate acid of NH_3. [1]

iii. Write the suitable example to show water acts as Bronsted-Lowry acid and base. [1]

OR

a. Rate of chemical reactions are affected by various factors.

i. How does temperature affect the rate of reaction? [1]

ii. With the energy profile diagram discuss the effect of catalyst in the rate of reaction. [2]

iii. Differentiate between homogeneous and heterogeneous catalysis with suitable examples? [2]

b. The free energy is a state function and depends on temperature, pressure and concentrations. Derive the relation $\Delta G = -T\Delta S_{total}$. Discuss the criteria of spontaneity on the basis of the above relation. [3]

10. a. The given table shows the different organic compounds.

A ⇒ $CH_3\ CO\ NH_2$	D ⇒ $CH_3\ COOH$
B ⇒ $CH_3\ CH_2\ OH$	E ⇒ CHI_3
C ⇒ C_2H_2	F ⇒ $CH_3\ CH_2\ NH_2$

Make a correct reaction sequence of the above compounds and proper conditions and reagent. [6]

b. Compare the acidic strength of formic acid, acetic acid and chloro acetic acid with reason. [2]

OR

a. Starting from methyl magnesium bromide (CH_3MgBr), how would you prepare

i. Propan-2-ol

ii. Ethanoic acid

iii. Methane [3]

b. Why does phenol give ortho and para substituted product during electrophilic substitution reaction? [2]

c. $C_6H_5OH \xrightarrow{A} C_6H_6 \xrightarrow{B} C_6H_5CH_3 \xrightarrow{C} C_6H_5CHO$

Choose correct reagent (A), (B) and (C) in the above reaction sequence. Write the IUPAC names of compounds given. [3]

11. a. What are azo- dyes? Write the structure of any one azo-dye. [2]

b. i. Differentiate between OPC and PPC cement. [2]

ii. Define the term clinker [1]

c. Distinguish between controlled nuclear fission and uncontrolled nuclear fission. [2]

d. What are the qualities of paper? [1]

- The End -